

Introduction

In this project we modeled the NO₂ concentration in the air over the past 30 years. NO₂ is a short-lived pollutant which primarily originates from fossil fuel burning. While it itself is not a major long-term greenhouse gas, like CO₂, its measurements provide good estimation for local air quality and climate co-benefits. It is highly toxic and the measurements are used to monitor air quality to protect public health as it is a key precursor to harmful ground-level ozone matter.

To provide some insight in the air quality over the past few decades and potentially infer what it will look like in the future, we used several Bayesian models to model the NO₂ concentration, specifically 3 variations of harmonic regression. Next, we evaluated the models using LOOIC, computed the AKAIKE weights for an ensemble and used the best model per LOOIC to quantify the egeneral NO₂ trend over time.

Methods

Data

The dataset has one dependant and one target variable, date and NO₂ concentration, respectively. Essentially the dataset records the NO₂ concentration for each day from 17.7.1997 to 9.12.2025. The target variable is continuous, ranging from 3 to 250 units. In total that's 10373 data points. The NO₂ concentration distribution is shown in black on Figure 2.

We performed a Fourier transform to identify dominant frequencies in the data. A strong peak at approximately 52 cycles per year indicates a weekly seasonal pattern, since $\frac{365}{52} = 7$ days. We therefore included a weekly seasonal component in the model. In addition, we incorporated a yearly seasonality to capture longer-term variation in NO₂ concentrations (e.g., increased emissions during winter heating periods). Before modeling, the time scale was centered and scaled to the interval $[-1, 1]$.

Models

We implemented 3 harmonic regression models:

Simple harmonic regression

Simple harmonic regression: We only model 2 seasonalities - weekly and yearly. The seasonality component is defined as

$$ssn_i = \sum_{j=1}^k (\beta_{\cos,j} \cos(\omega_j t_i) + \beta_{\sin,j} \sin(\omega_j t_i))$$

and the linear component (trend + seasonality) as

$$y[i] \sim N(\alpha + \beta * t[i] + ssn_i, \sigma^2)$$

where N is the number of data points, t the time step and ω the frequency. To keep the model simple, each seasonality is represented by a single sinusoidal term, corresponding to one full cycle per period.

The intercept and other priors were set to $\alpha \sim N(90, 10)$, $\beta \sim N(0, 1)$ and the seasonality coefficients were left as the default priors.

Harmonic regression with complex seasonalities

To investigate whether modeling more complex sinusoids improves performance and allows the model to capture the NO₂ concentration variability better we also built a harmonic regression model with the same ω values as the simple model, except we combined 2 and 5 sinusoids for weekly and yearly seasonalities respectively. The model

definition is similar, but now we have multiple $\beta_{\sin}$ and $\beta_{\cos}$ parameters, allowing us to combine several sinusoids to model a cycle. Specifically, our seasonality component is defined as

$$ssn_i = \sum_{s=1}^S \sum_{j=1}^{k_s} (\beta_{\cos,s,j} \cos(\omega_{s,j} t_i) + \beta_{\sin,s,j} \sin(\omega_{s,j} t_i))$$

where $k_{\text{week}} = 2$ and $k_{\text{year}} = 5$. The trend priors were the same as for the simple regression, however the $\beta_{\sin}$ and $\beta_{\cos}$ parameters were set to $N(0, 1)$ to speed up the sampling process.

Harmonic regression with quadratic trend

Lastly, we focused on modeling the trend better, since that seemed to be the biggest shortcoming of the previous models. In order to do so, we modeled the trend with a polynomial of the second degree (quadratic function). To contrast whether this approach is better than just focusing on the seasonalities we modeled those the same as in the first example (simple harmonic regression). The seasonality is therefore the same as for the simple model,

$$ssn_i = \sum_{j=1}^k \beta_{\cos}[j] * \cos(\omega[j] * t[i]) + \beta_{\sin}[j] * \sin(\omega[j] * t[i]),$$

and the final model is defined as

$$y_i \sim N(\alpha + \beta_{lin} * t[i] + \beta_{poly} * t[i]^2 + ssn_i, \sigma^2).$$

For this model, we kept the priors the same, however since now we had a new parameter β_{poly} which we set to $N(0, 1)$ as well. In retrospect, the selection of priors could be more sophisticated; however, given that we have enough data, the final result should not deviate significantly if other values were chosen.

Results

General insights

Figure 2 shows the posterior means with 95% HDIs plotted over the original data. Notice that the posteriors differ quite significantly – the simple model has a very obvious simple yearly seasonality (the weekly seasonality can't be seen visually at this scale), and the trend has a slope of approximately 0. Even visually, we can determine that this is not a very good fit, and the main reason for that is most likely that the model was influenced by the high leverage points (outliers). Considering this, a Cauchy or a Student distribution would be better to model this trend, since it's more robust to outliers.

Taking a look at the multi-seasonal model, we immediately notice that the trend looks more accurate now. The reason for this might be that the seasonalities were able to better capture and adjust for the outliers, and allowed the trend to fit better. Specifically the yearly trend, made up of a sum of 5 sinusoids looks relatively complex and further increasing the number of parameters for modeling it might cause overfitting.

Lastly, the harmonic regression with a polynomial trend seems to have fit the data the best. The trend looks reasonable and the yearly seasonality is not as prominent as it was in the simple model.

Our models were able to fit the data well, however keep in mind that none of them are suitable for long-term extrapolation.

Model	Δ ELPD	Δ SE
QT	0.0	0.0
Multiseasonal reg.	-679.3	99.8
Simple harmonic regression	-7435.5	153.7

Table 1. LOOIC score comparison among the models. The differences between the models are pretty extreme, indicating that the QT model is significantly better than the other two.

LOOIC evaluation

To obtain a more accurate measure of model performance, we used the Leave-One-Out Information Criterion (LOOIC), which estimates out-of-sample predictive accuracy by approximating leave-one-out cross-validation using Pareto-smoothed importance sampling. LOOIC provides an estimate of how well each model is expected to predict new, unseen observations, penalizing overly complex models that do not generalize well. To get LOOIC scores we computed the log likelihoods for all points for each model and used the LOO function in R. The results are shown in Figure 1. Table 1 shows the LOOIC score comparison between the models, and shows that the QT (quadratic trend) model significantly outperforms the other two.

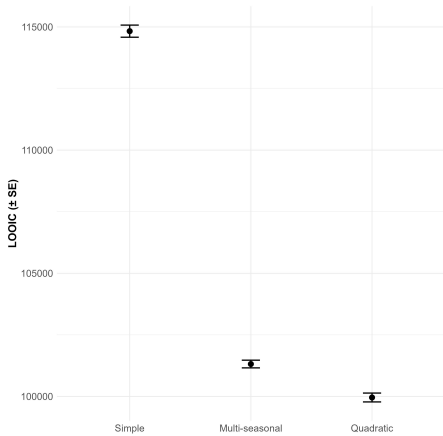


Figure 1. LOOIC + SE scores for each models. Notice that the model with QT significantly outperforms the other two, with the simple model being the worst by far.

Model ensemble and Akaike weights

Since ensemble models often outperform other models we also computed the akaike weights for a model ensemble. This way, we should be able to leverage the strengths of all the models to get a better prediction. Surprisingly, the Akaike weights for the simple and multiseasonal models were both 0 and only the QT model was used. The expected outcome would be that the ensemble would leverage both the QT and the multiseasonal component of the last two models, however it appears that the QT model is able to fit to the data way better than the other two models, therefore, only that model is picked.

Quantifying the general NO2 trend over time

Given that relative to the other models we developed, only the QT model is relevant, we also quantified the NO2 concentration trend over time with it. We computed the **average trend** over per day and per year and got the following results: The NO2 concentration falls approximately **0.0061 and -2.2192 units per day** and per year respectively (**HDI: [-0.0064 , -0.0057], [-2.3497, -2.0930]**). Figure 3 shows the mean pointwise trend per day across the entire dataset, calculated as the mean of the derivatives of the quadratic model. The strong negative trend is primarily driven by the high NO2 measurements between 2004 and 2010 compared to the much lower concentrations observed after

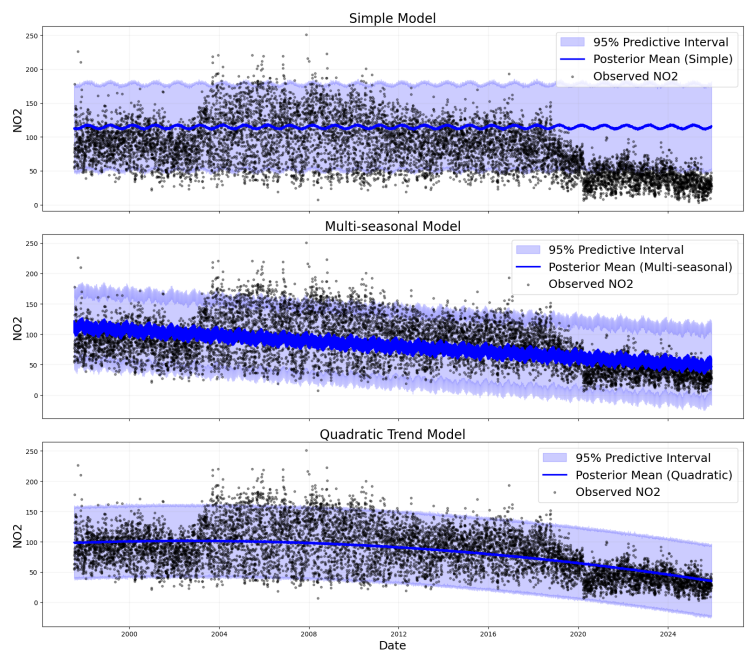


Figure 2. Raw data and each model's posterior mean and 95% HDIs. Notice the differences in the seasonality components and trends and how they affect the goodness of fit.

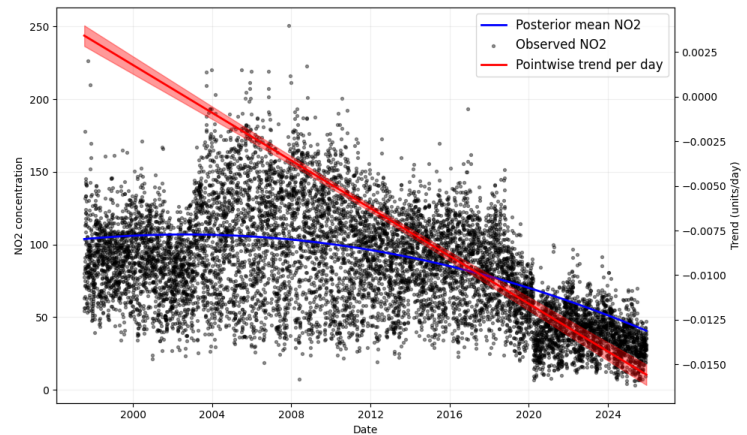


Figure 3. Mean trend over the entire dataset, 95% HDIs and the posterior mean over the original data.

2020. This also highlights why a linear model would struggle to capture the data patterns accurately, given the imbalance in NO2 concentration levels over time.

Keep in mind that while the QT model fit is relatively good, it should not be used for any extrapolation, as by its geometric property, the trend will keep decreasing. In fact, one should be careful when extrapolating anything when working with a time series data.

Conclusion

The analysis demonstrates that NO2 concentrations in the observed region have declined substantially over the past 30 years, with the strongest trend captured by a harmonic regression model with a quadratic trend. While simple harmonic regression fails to capture the nuances of seasonal variability and trends, and multi-seasonal models improve fit but risk overfitting, the quadratic trend model balances complexity and predictive accuracy, as confirmed by LOOIC evaluation. The decreasing trend in NO2 is consistent with reductions in high-emission periods post-2020. Further work can include combining our models with autoregressive components, moving averages, different distributions and other components based on potential domain knowledge to model the data better.